

HomeWork - Correlation&Convolution

Q1. Perform correlation and convolution operations on highlighted pixels of given image data.

image data

4	5	6	5	4	4
3	3	4	3	2	3
4	4	6	6	7	4
1	2	2	3	6	1
2	3	5	5	6	2
4	7	6	2	2	4
7	6	5	3	4	7

kernel 1 (calculate on red pixel)

2	4	4
4	3	4
2	3	3

kernel 2 (calculate on blue pixel)

3	3	3	2	2
4	3	3	2	1
4	4	3	2	2
5	4	2	1	0
4	3	1	0	0

Q2. For given image data perform zero padding, replication, wrap around(circular) and mirror padding.

2	2	5	6	7	4	4
3	3	4	4	5	6	4
1	2	7	7	5	5	2
5	5	6	6	4	4	3
1	1	2	2	7	7	6
4	4	5	5	7	7	6
4	5	6	6	7	4	3

i. pad image for 3*3 kernel

ii. Pad image for 5*5 kernel

Q3. Perform correlation and convolution operation on given image using pixel omitting and truncate the image method.

image

6	5	5
3	5	5
5	7	7
0	0	2

kernel

6	4	4
6	2	2
4	4	2

Q4. Perform correlation and convolution operation on pixels highlighted in the image with suitable padding.

use kernel 1 and kernel 2 from Q1 for red and blue pixels respectively.

4	5	6	5	4	5	4
3	3	4	3	3	3	2
4	4	3	6	4	6	7
1	2	2	6	1	6	6
2	3	5	5	2	5	6
4	7	6	2	4	2	2
7	6	3	5	7	3	4